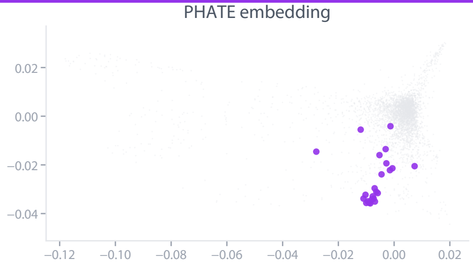
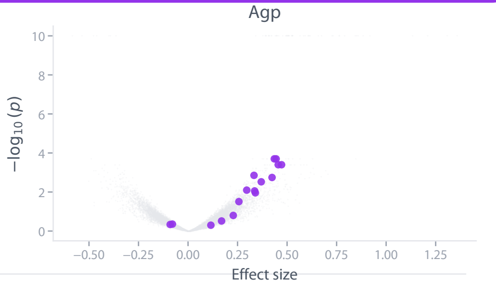
RNA Polymerase II transcription and elongation

Cluster 8





Summary

This cluster shows HIGH confidence for RNA Polymerase II transcription and elongation as the dominant pathway. 17/23 genes (74%) are established components: 10 Pol II core subunits, the general transcription factor GTF2B, the PAF1 complex (PAF1, CTR9, CDC73), the elongation factor SUPT5H, the CTD phosphatase CTDP1, and the CCR4-NOT deadenylation complex scaffold CNOT1. The morphological phenotypes...

Established genes

`POLR2B, POLR2H, POLR2C, POLR2J, POLR2J2, POLR2J3, POLR2F, POLR2L, POLR2K, POLR2E, GTF2B, PAF1, CTR9, CDC73, SUPT5H, CTDP1, CNOT1`

Novel & uncharacterized genes

DDX3X *novel*

Multifunctional RNA helicase with documented roles in transcription regulation (CDKN1A, CDH1, HNF4A-mediated transcription). However, its clustering specifically with Pol II elongation machinery

SENP6 *novel*

SUMO protease with established roles in chromosome segregation and DNA repair. While SUMOylation regulates transcription factors, SENP6's clustering with Pol II machinery suggests potential direct

UBE3A *novel*

E3 ubiquitin ligase with established roles in protein degradation and quality control. While it has transcriptional coactivator activity with progesterone receptor, its clustering with core Pol II

RHOC *novel*

Small GTPase known for regulating cytoskeletal dynamics, focal adhesions, and cytokinesis. No documented role in transcription. Its presence in this cluster suggests potential involvement in

HMGCR *novel*

Rate-limiting enzyme in cholesterol biosynthesis with well-established metabolic functions. No known direct role in transcription. Clustering with Pol II machinery suggests potential novel connection

PLPP4 *novel*

Magnesium-independent phospholipid phosphatase with no established connection to transcription machinery. Its clustering with core Pol II components suggests a potential novel role in

17 established · 6 novel · 0 uncharacterized